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Land cover mapping for tropical forest rehabilitation planning using remotely-sensed data

A. A. APAN

Department of Geography and Environmental Science, Monash University,
Clayton, Victoria 3168, Australia

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Abstract. Lack of reliable and up-to-date maps relating to land cover (among other themes) constitute a weakness in land resource surveys and cause costly failures to many forest rehabilitation projects in the tropics. This study evaluated the utility of satellite imagery for land cover mapping for forest rehabilitation planning in a case study in Mindoro, Philippines. Using Landsat TM data, visual and digital image processing techniques were performed using the GRID module of ARC/INFO and the microBRIAN image processing software. Crown cover density is found as the most useful and the most important detail of information the image could provide. Detailed mapping at the species and forest type levels is unreliable, as is the delineation of water bodies and some cultural features in rugged terrain. Clustering of the NDVI image is found more applicable in producing land cover maps depicting crown cover classes than classifying raw TM-3, -4, and -5.

1. Introduction

Tropical forest destruction is relatively uncomplicated. In contrast, reforestation millions of hectares of mostly eroded, unproductive, inaccessible and rugged deforested areas is not an easy job. The tropic's unique bio-physical attributes make forest rehabilitation increasingly difficult. Moreover, the dynamism of various socio-economic driving forces, especially in developing countries, contribute to the complexity of any rehabilitation effort. As a result, many forest rehabilitation programs and projects fail; while others record only lackluster performances (e.g., Krochmal and Krochmal 1986, pp. 65–78, Jackson 1984).

One of the main causes of forestation failures refers to poor spatial planning and monitoring of projects, due to weakness in land resource surveys and evaluation systems (e.g., Tolentino 1992, Apan 1992). This is particularly true for most developing countries because spatial data sets are often lacking. Sound decision-making during planning is often hampered by lack of reliable and up-to-date maps. Land cover information is an important theme because it affects more of the important decisions that have to be made during forest rehabilitation planning than do most other themes.

Remote sensing is a powerful tool that could be used to address the problem of out of date thematic maps. The capabilities of remote sensing to map and extract information about earth resources for various applications are well documented. Among those prominently used is land cover mapping, considered as one of the most important, most direct and well established applications of remote sensing (Campbell 1987, p. 497). However, the classification of land cover with satellite data

is not a perfect process as many sites are often misclassified (Czaplewski 1991). In addition, there is no standard approach in extracting land cover information that could be applied for all situations.

The general objective of this study is to evaluate the utility of satellite imagery for land cover mapping for forest rehabilitation planning in Mindoro, Philippines. The following are the specific objectives: (1) to determine the details of land cover information that could be accurately mapped for forest rehabilitation planning using Landsat TM data, (2) to identify some image processing techniques that could be routinely used, and (3) to produce a land cover map from the classification of Landsat TM data. It is a part of an integrated GIS-based study on spatial planning of forest rehabilitation projects in the Philippines.

2. The need for land cover information

In their recent study in the Philippines, consultants of the U.S.-based Development Alternatives, Inc. identified the minimum data sets needed for developing a spatial information system for natural resources management (DAI 1993). First in the list, land cover information is identified as very important (first priority) because it provides a major portion of the information needed for natural resources planning. An accurate knowledge of land cover features and their extent represents the foundation for land classification and management (DAI 1993). The availability of accurate land cover information is an important element in making sound decisions concerning Earth's resources and the environment.

With regards to forest rehabilitation planning, land cover information are even more than usually necessary. In their example of spatial analysis for reforestation, DAI (1993) identified land cover as one of the priority data sets needed. Information on land cover are required in some of the following relevant forest rehabilitation tasks: (1) identification of plantable and conservation areas (Evans 1992, p. 173), (2) formulation of management objectives, (3) identification and design of forest rehabilitation or silvicultural treatments (JOFCA/ITTO 1993), (4) analysis of landscape patterns and structure (Apan 1996a), (5) infrastructure design, (6) resource prioritization and allocation (Apan 1996b), (7) forest protection and maintenance, (8) project monitoring and evaluation, and (9) costs and financial planning.

The latest existing set of land cover maps in the Philippines is a product of a nation-wide mapping project that used 1987–1988 SPOT imagery (Swedish Space Corporation 1988). The objective of the project was to satisfy the World Bank's need for up-to-date information on the land cover of the Philippines for their study of the country's resources. However, the utility of the above data for provincial or local-level planning is inadequate with regards to spatial scale. The map series produced from the SPOT data are in 1:250 000 scale, and the smallest mapping unit was 100 ha. Furthermore, these maps are also considered outdated for present-day planning since the pattern of land cover features, especially vegetation, is highly dynamic.

3. What detail of land cover information

Land cover mapping requires that surface features be classified into various categories. The level or detail is dependent on many factors: the objectives of mapping, the information need, the spatial and radiometric resolutions of the sensor, the environmental condition of the target area, the analytical techniques to be applied, and others. For forest rehabilitation planning, a basic information require-

ment is the knowledge concerning the boundary between vegetated and non-vegetated areas (DENR 1988, 1991, Evans 1992, p. 173). This primarily determines the location of potential rehabilitation areas. On the other hand, more detailed information are required for other decisions affecting the objective of management, the silvicultural treatments, and the prioritization of areas, among other things. These include the kind and condition of vegetation, the presence of road and trail, the presence of human settlements, and the location of water bodies (e.g., Evans 1992, p. 173, JOFCA/ITTO 1993).

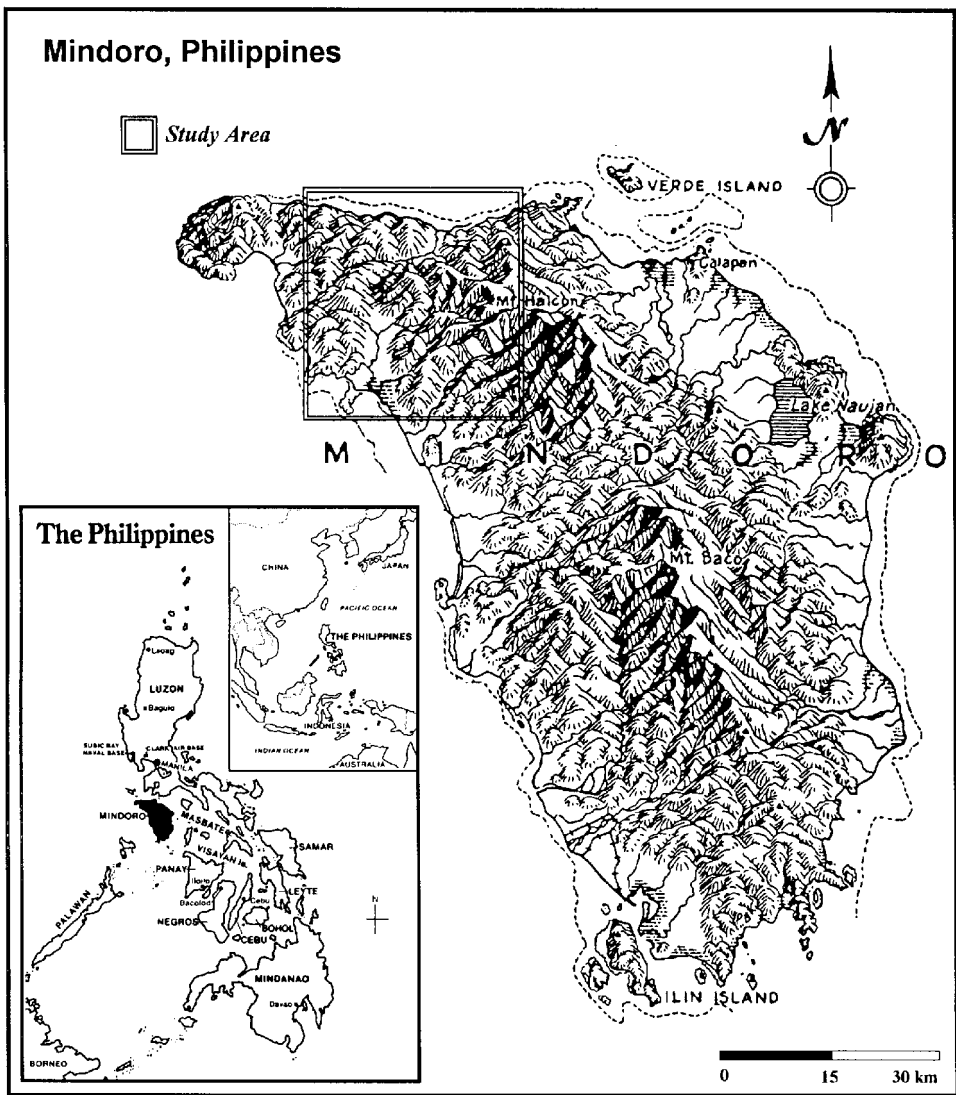
A number of studies noted the difficulty in separating the spectral values of tropical vegetation types, and their subsequent classification. Sader *et al.* (1990) noted that old secondary forest and disturbed primary forest could not be distinguished from each other, and early successional stages were often confused with second growth forest and regions of mixed crops. In a study in southeastern Mexico, Garcia and Alvarez (1994) recognized the difficulty in separating the brightness values of secondary vegetation classes using TM data. Lucas *et al.* (1993) also noted the difficulty in discriminating plantations from regenerating secondary forest, when they used multi-temporal Landsat imagery. Similar difficulties were identified by Apan and Peterson (1994) in a spectral discrimination study of forests and plantation areas in the Philippines using TM imagery.

However, mapping of tropical forests at broad level seems possible using satellite data. Mausel *et al.* (1993), in their study of an Amazon area, were able to classify, with 82–88 per cent accuracy, three classes of secondary succession forests (initial, intermediate, advanced) using TM images. Palubinskas *et al.* (1995) found that texture-based classification and post-classification of fuzzy classification output yielded high accuracy in mapping regenerating tropical forest classes of a region in Amazonas, Brazil using TM data. In some studies in the tropics, high classification accuracies were achieved even in detailed and semi-detailed mapping (i.e., species of forest type levels) of vegetation and other land cover types (e.g., Roy *et al.* 1991, Saxena *et al.* 1992).

4. Study area

The study area is located in the province of Occidental Mindoro, Philippines, approximately 120 km south of Manila (figure 1). It covers the municipalities of Abra de Ilog and Mamburao, with an area of about 84 795 ha. The area's topography is generally mountainous and rugged; dissected by many rivers and creeks. Elevation ranges from sea level to 1709 m above mean sea level. The forest zones, of varying successional classes, are mainly composed of the dipterocarps and molave types: the former is dominated by lauan (*Shorea contorta*) and apitong (*Dipterocarpus philippinensis*), while the latter is dominated by molave (*Vitex parviflora*), narra (*Pterocarpus indicus*) and ipil (*Intsia bijuga*). Mangrove forests are also present, on lowlands often converted for fishponds and other land uses. The grassland areas, often subjected to forest fire, are mainly colonized by *Imperata cylindrica*. The province has two pronounced seasons: dry from November to April, and wet during the rest of the year. Monthly average temperature ranges from 26.8°C to 31.7°C.

National Statistics Office (1992) census showed that the two municipalities under study have a total population of about 35 390 as of 1990. Most of the people work as farmers, fishermen, and laborers. Primary agricultural produce includes rice, coconuts, bananas, fruits, and vegetables. Towns and villages are mostly concentrated in the coastal and alluvial plains, although some inhabitants dwell on lower and



Adapted from *Allied Geographical Section (1944)*

Figure 1. Location map of the study area.

midslope portions of mountains. The province has indigenous inhabitants, the Mangyans. It is also the home of the endangered tamaraw (*Bubalus mindorensis*), a buffalo-like upland animal that can be found only in Mindoro.

Licensed commercial logging in the area started in the late 1960s and ended in 1983. In 1987, there were about 40 pasture lease agreement holders covering some 23 825 ha. Permits to utilize minor forest products such as charcoal, rattan, palm shingles and midribs, and beeswax, were likewise issued. Currently, a commercial logging ban is enforced in the whole province. In the late 1970s, two major government reforestation projects were established. In the early 1990s, the launching of the National Forestation Program dramatically increased the establishment of forestation projects.

5. Methods

5.1. Introduction

Four details of land cover attributes were investigated: (1) forest types, (2) crown cover or canopy closure, (3) successional stages, and (4) non-vegetated areas. Both visual and digital image analysis approaches were employed (figure 2). The first stage of image analysis and some aspects of clustering were performed using the PC-based microBRIAN Ver. 3.1 Image Processing System (MPA 1992). When become available and found more appropriate, the rest of the processing was done using the GRID module of the Unix-based ARC/INFO Revision 7 (ESRI 1995) and IDRISI for Windows (Eastman 1995). Hardcopy outputs were plotted using the Mitsubishi G650 thermal colour printer and Tektronix 4696 colour inkjet plotter.

5.2. Dataset acquisition and field data gathering

A 1420 pixel $\times$ 1350 pixel subset was used from a Landsat Thematic Mapper digital data (Path/Row 116-51), taken 11 February 1992 at 1:40 p.m. All the visible and infrared bands, except the thermal infrared, was included in the analysis. The

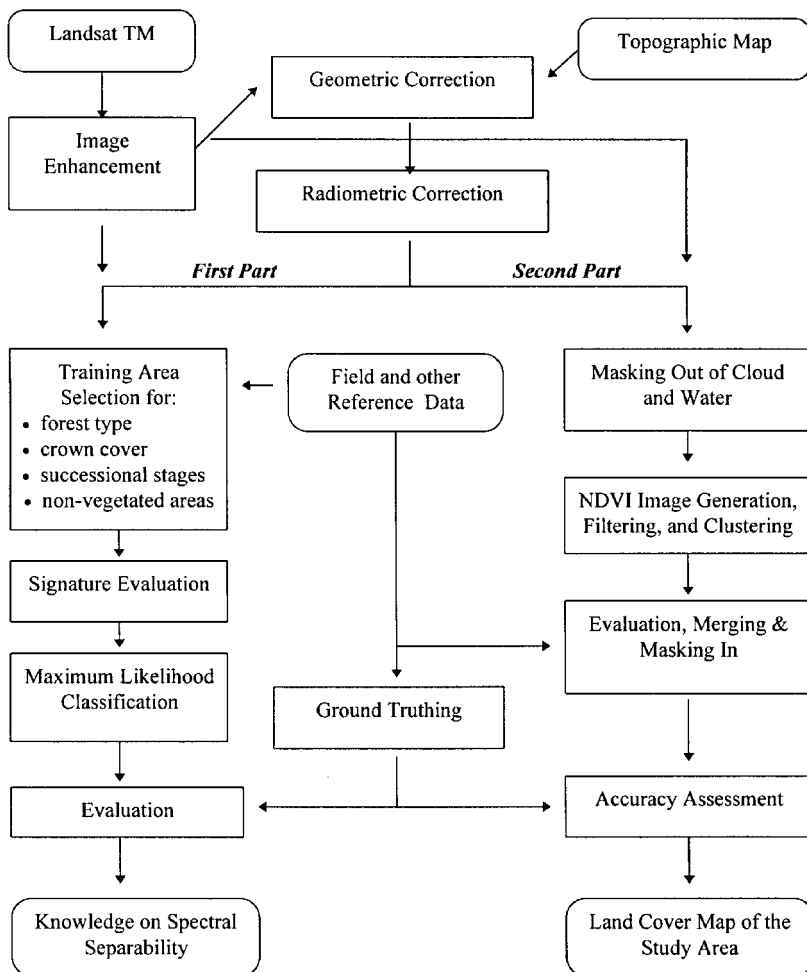


Figure 2. Schematic diagram of the image analysis method.

acquisition data falls within the area's dry season. It was envisioned that another image scene corresponding to the wet season might be helpful in the analysis. However, there was no available data for the wet season. Aside from the resource constraints of obtaining one, acquiring a sufficient cloud-free image during wet season in the tropics is a major problem. Thus, this option was not pursued.

The ancillary data used in this study included the following: topographic map, land use and vegetation map, landform map, and soil map. They were all acquired in 1 : 50 000 scale (except the 1 : 200 000 scale soil map) from the Philippines' National Mapping and Resource Information Authority and Bureau of Soils and Water Management. Field reconnaissance survey was carried out in the initial stage, while intensive surveys were completed during the middle and final stages of image processing.

Crown cover is *the proportion of a unit area of ground covered by a vertical projection of tree crowns* (Philip 1994, p. 132). Its actual field measurement is difficult and laborious. In this study, crown cover is determined by the consensus of the researcher and two professional foresters aided by crown density scale proposed by Deutsche Forst-Consult (1992). Three categories were defined: low crown cover (<10 per cent), medium crown cover (10–40 per cent), and high crown cover (>40 per cent) (terminology modified from Singh 1989, but similar in percentage ranges). Successional stages were identified and described based on the age and structure of vegetation to come up with three classes: initial, intermediate, and advanced. On the other hand, identification of forest types is based on dominant species, which is relatively straightforward.

5.3. Visual image analysis

The primary goal of visual image analysis was to examine whether the details of land cover attributes could be visually identified and delineated using Landsat TM data. Visual interpretation also helped in the identification of land cover features needed in (1) the identification of ground control points for geometric correction (2) the selection and evaluation of training samples, (3) the masking out of cloud and water, (4) the evaluation and merging of clusters produced from unsupervised classification, and (5) accuracy assessment of the classified image.

The visual interpretation of these images was greatly facilitated by image enhancement techniques. These include contrast enhancements using linear and histogram-equalization stretching. Different colour composites were produced (1 : 50 000 and 250 000), to include the following: colour composite of TM-3, -4, and -5, colour composite of TM-1, -2, and -3, and the first three principal component transformation images of all TM bands, except TM-6.

5.4. Digital image analysis

5.4.1. Geometric and radiometric corrections

Using microBRIAN software, the image was geometrically rectified to 1 : 50 000 scale topographic maps projected in transverse mercator. Twenty-one ground control points were located for both the image and maps. Using a first order affine transformation function and nearest neighbour resampling algorithm, the image was rectified attaining an allowable root mean square error of about 24 m and 26 m for x, y directions, respectively. The original 30 m × 30 pixel size was retained.

A microBRIAN destriping program was used to minimize the striping effect of the sensor. For atmospheric correction, a haze removal technique based on the dark

value (water bodies/shadows) principles (Richards 1993, pp. 46–47) was carried out. Detailed correction for the effects of the atmosphere requires modelling of the complex processes that took place, and establishing how these determine the transmittances of the various paths and the different components of sky irradiance and path radiance (Richards 1993, p. 44). This requires reliable data such as visibility and relative humidity. These data are not available for the study area, thus detailed correction could not be performed. However, the above technique is considered adequate, mainly because of the seemingly even spatial distribution of atmospheric haze as seen from the enhanced TM-1 and -2 images.

5.4.2. *Correction of topographic effects*

Correction methods for the impact of topography on the radiometry have met with marginal success to date, and a widely applicable procedure is still wanting. Most methods using data extracted from the digital elevation models (DEMs) are still unreliable. Even the Lambertian assumption where most DEM-based correction techniques are based, carries inherent limitations (Smith *et al.* 1980). Moreover, most DEM construction relies on interpolation techniques which are also highly prone to errors. For example, resulting images tended to be over-corrected, with slopes facing away from the Sun (i.e., northerly-facing slopes in the northern hemisphere) appearing brighter than sun-facing slopes of similar cover composition (Kawata *et al.* 1988). In a study of tropical forest regeneration in Ghana, Curran and Foody (1994) met some difficulty with the DEM approach, and they discontinued its use in the correction of remotely-sensed radiance in that particular study.

With these considerations, DEM-based approach such as the cosine correction method was not used in this research. Instead, the normalized difference vegetation index (NDVI) image $((\text{TM-4} - \text{TM-3})/(\text{TM-4} + \text{TM-3}))$ was produced in hoping to minimize topographic effects. Aside from being a popular vegetation index image, NDVI could reduce the topographic shadow effect in high relief areas (Lillesand and Kiefer 1987, pp. 650–655). However, it is certainly not safe to assume that the topographic effects is completely removed by a ratio operation, such as NDVI (Mather 1987, p. 198).

The expected range of the NDVI values is from -1 to $+1$. A function to transform the range of the raw NDVI values to 0–255 scale was derived by adding 1.0 to the raw NDVI value (making the range 0–2) and multiplying the result by 127 (Mather 1987, p. 198). This scaling is necessary to convert the ratio values to integers on the range 0–255 for display and classification purposes. Since ratio operations tend to generate noise, a low-pass 3 by 3 median filter was applied.

5.4.3. *Training area selection and signature evaluation*

Training areas for each of the four land cover attributes under study were selected based on *a priori* knowledge, and information from ancillary maps and field data gathering (tables 1–4). The program allowed on-screen digitizing and editing of training samples from the display of various enhanced image composites. Samples were located in areas apparently not affected by topographic shadows. Four signature files containing statistics for the different land cover attributes were generated. The statistics contained in the signature file aid in determining whether the classes are good (no or little overlap) in terms of their separability in the multi-dimensional attribute space. As provided by the software, four graphical tools to explore the statistical distribution of the classes are used: histogram, scattergrams, ellipses, and dendrograms.

Table 1. Identification and description of training samples for forest type.

Training area class	No. of pixel	Description	Appearance in TM-3, -4, and -5 (BGR) composite
Molave-dipterocarp	267	Mixture of molave and dipterocarp forest types species (e.g. <i>Pterocarpus indicus</i> , <i>Vitex parviflora</i> , <i>Dracontomelon dao</i> , <i>Instia bijuga</i>), closed canopy forest, advanced successional stage	Dark green
Shrub	152	Mixture of dense grasses (mainly <i>Imperata cylindrica</i>), palms (e.g. <i>Arenga pinnata</i>), shrubs (e.g. <i>Ervatamia</i> sp., <i>Musa</i> sp.), and trees (e.g. <i>Trema orientalis</i> , <i>Ficus</i> and <i>Artocarpus</i> sp., <i>Syzygium cumini</i> , and <i>Psidium guajava</i>)	Yellowish green
Grass	156	Mostly dried grasses of <i>Imperata cylindrica</i> , <i>Saccharum spontaneum</i> , and <i>Themeda triandra</i>	Whitish to light magenta/pink
Mangrove	313	Mature mangrove vegetation: mainly <i>Rhizophora</i> , <i>Avicennia</i> , and <i>Bruigeria</i> spp.	Dark green
Tree plantation	75	Mature, dense tree plantation planted with <i>Gmelina arborea</i> , <i>Swietenia macrophylla</i> , and <i>Leucaena leucocephala</i>	Dark green
Coconut	156	Mature, dense coconut (<i>Cocos nucifera</i>) plantation, mostly with fruit trees and other crops underneath	Dark green
Mixed agri crops	203	Mixture of corn, rice, green leafy vegetables, bananas, etc.	Green (less darker)

Table 2. Identification and description of training samples for crown cover attribute.

Training area class	No. of pixel	Description	Appearance in TM-3, -4, and -5 (BGR) composite
Low	130	Bare soil and eroded areas, grasses, cultivated areas, low crown cover (<10%)	Whitish to magenta
Medium	141	Mixture of shrubs, grasses, and tree regeneration, medium crown cover (10–49%)	Light, yellowish green
High	138	Dense forest, mature trees, mixed molave and dipterocarp type forests, high to closed crown cover (50–100%)	Dark green

5.4.4. Supervised classification and evaluation of results

The goal of supervised classification in this part is not to produce a land cover map. The presence of topographic shadows in high relief areas is expected to yield substantial classification errors. Instead, the conduct of classification was to determine how the various training samples will perform, and to locate where confusions have emanated. Analysing the separability of training samples will indicate the detail or level of land cover information that could be mapped for forest rehabilitation purposes.

Table 3. Identification and description of training samples for successional stages attribute.

Training area class	No. of pixels	Description	Appearance in TM-3, -4, and -5 (BGR) composite
Initial	258	Bare soil and grasses, no developed vegetation communities	Whitish to magenta
Intermediate	162	Young forest formation, developing and partially open canopy, mixture of young and mature trees, developing forest structure	Light, yellowish green
Advanced	173	Mature forest formation, developed and closed canopy, mature trees, more complex forest structure than intermediate	Dark green

Table 4. Identification and description of training sample classes for non-vegetated areas.

Training class	No. of pixels	Description	Appearance in TM-3, -4, and -5 (BGR) composite
Bare soil1	308	Dry agricultural lands (crops harvested)	Magenta/pink
Bare soil2	129	Upland bare soil and highly eroded areas	Dark magenta/pink to violet
Built-up	135	Towns of Mamburao and Abra de Ilog: mainly roads, buildings, parks, galvanized iron houses, trees, and hut made of dried <i>Nypa</i> palm leaves	Bluish and magenta
Water	67	Lake, river, fishponds, and irrigated areas	Black
Shadow	48	Topographic shadows in high relief areas	Black

The supervised classification in this study utilized the maximum likelihood algorithm. The signature files generated from the above procedure were used, with all classes set to equal probability. Every cell was set for classification, i.e., without setting any reject fraction—the portion of cells that will remain unclassified due to lowest possibility of correct assignment (ESRI 1995). The TM bands used were -3, -4, and -5. These bands (including band 7) were found to yield highest classification accuracy (Peterson *et al.* 1986), and they are the best TM bands for general forest cover-type mapping (Horler and Ahern 1986). In a tropical forest type mapping study, Roy *et al.* (1991) also favoured these bands in their image classification.

At this stage, the evaluation of the classification results was done by a less thorough visual comparison of the classified images and ancillary data. Samples which are not apparently affected by topographic shading were purposely selected and compared with the ground truth. Area statistics of the various classes for the classified images were also verified in terms of their validity. For example, it is not valid to yield a total area of more than 800 ha for water bodies, or more than 1000 ha of mangrove forest in the study area.

5.4.5. Masking, unsupervised classification, and merging

The microBRIAN's clustering program, a nearest neighbour classifier with optional migration of class means, was implemented to the NDVI image. Before

this, however, small areas covered by clouds and water were digitized out by visual interpretation. Cloud cover conceals the corresponding surface features. Water, on the other hand, is expected to yield values similar or very close to non-vegetated features in ratio images (e.g., Campbell 1987, p. 385). Dissimilar materials with different absolute radiances, but having similar slopes on their spectral-reflectivity curves, may be inseparable or may appear identical in ratio images (Lillesand and Kiefer 1987, p. 654).

In classifying, a number of trials with varying combinations of parameters were tried. Finally, the number of classes was set to the program maximum (100 classes), the gate tolerance (the absolute bounds within which a pixel must fall to be a member of the class) to minimum (1 per cent of the channel range), and the number of iterations to 2. The results of the classification were further analysed using a program that produced a minimum spanning tree plot (a set of links joining classes which are close to each other). The identity of spectral classes was carefully determined by selective display in the video monitor, and their comparison with reference and field data. Then, merging of these spectral classes into information classes was made cautiously. In addition, the layer containing water features that was previously masked out was combined to the classified image.

5.4.6. Accuracy assessment

The accuracy of the results of NDVI clustering was evaluated through comparison with ground reference data. Two hundred and eighty-seven (287) reference pixels were generated using simple random sampling by a GRID program. This sampling scheme was selected because it can provide adequate estimates of the population parameters, provided the sample size is sufficient (Congalton 1988). About 76 per cent of samples (218 pixels) were actually inspected on the ground, while the rest (because of time and accessibility constraints) are either based from (1) the 1988 photo-interpreted land use and vegetation maps, and/or from (2) field knowledge of local foresters. Error matrix was generated using a program in IDRISI.

6. Results and discussion

6.1. Visual image analysis

6.1.1. Vegetated-bareland features

As expected, visual image interpretation of enhanced Landsat-TM data allowed the easy differentiation between the vegetated and non-vegetated areas. The contrasting reflectance properties of bare areas and vegetation in the visible and infrared bands, allow their easy differentiation. In TM-3, -4 and -5, vegetation appears in varying tones of green and barelands in pink to magenta. Bareland areas have high brightness values in the visible bands and low brightness values in the infrared band, while the reverse is true for vegetation. The above differentiation is more unrestrained using TM-3, -4, and -5 in a BGR colour composite (contrast stretched using histogram equalization technique). TM-1, -2, and -3 colour composite also offered good differentiation. These band combinations are consistent with the general findings (e.g., Ahern and Sirois 1989, Roy *et al.* 1989).

This study realizes that the uncomplicated delineation of vegetated and bareland areas from visual interpretation of Landsat data is probably the most dependable in terms of accuracy, and the less complex in extracting land cover information for forest rehabilitation purposes. Because vegetation-bareland delineation is a required basic information for forestation, the use of remotely-sensed data for the

purpose will be of great value. The location and prioritization of candidate areas for rehabilitation and the estimation of their extent could be easily done.

6.1.2. *Forest type classes*

With regards to forest type classes (e.g., molave-dipterocarp forest and mature tree plantation), accurate differentiation is difficult, except for mangrove species which could be associated with river-sea interfaces and deltas. This is most likely due to the following reasons: (1) the canopy structure of the major forest types are not significantly different, (2) the tree plantations are not in large contiguous areas, i.e., they are in irregular, relatively small patches which are interspersed with forested areas, (3) there is no uniform spacing and species composition of planted trees in most plantations, and (4) most coconut plantations are interspersed with fruit trees, thus their canopies mimic natural forests. These could partly explain why the 1987 SPOT-based land cover mapping project in the Philippines has committed the error of indicating coconut plantations (intensively intercropped with fruit trees) as closed forest on the lower slopes of Mount Banahaw (DAI 1992).

6.1.3. *Crown cover and successional stages*

The identification of different crown cover classes as well as successional stage classes could be inferred from varying tones of green. However, their exact boundaries are difficult to delineate. In TM-3, -4, and -5 (BGR), high crown cover vegetation appears as green to dark green, while low crown cover vegetation appears as light green to yellow-green.

While forest type discrimination seems impossible, classes of vegetation crown cover and successional stages are more pronounced, thus offer more potential use for land cover mapping for forest rehabilitation. High crown cover density and low crown cover density vegetation could be located in the image although they could not be sharply delineated from the intermediate medium crown cover class. This is possibly because the human eye is limited in detecting subtle differences in the spectral pattern of gradual change in density or successional stage classes. In this situation, the imagery's use is confined to the overall appreciation of the differences of crown cover density or successional stage classes and their approximate location, but utilizing it for mapping in the context of planning and monitoring purposes will be less reliable. In any case, the use of TM-3, -4 and -5 colour composite is most useful, and this agrees with Ahern and Sirois (1989), Saxena *et al.* (1992) and Roy *et al.* (1991).

6.1.4. *Cultural and non-vegetated features*

Cultural features such as roads and trails, houses, buildings, etc. have low separability over bareland features. Roads are more visible in TM-1, -2, and -3 (BGR) than any other composites, because of their linear patterns rather than its colour. Trails were neither identified nor found on any of the images. Roads could not be well discriminated in built-up areas or under thick vegetation. Water bodies could be adequately discriminated, but not by its black to bluish colour in TM-3, -4, and -5, but by pattern and associated location. Most topographic shadows, being black, can be confused with deep water in classification.

Therefore, under similar conditions prevailing in the study area, Landsat TM data's utility for the location of road/trail network and settlement relevant in forest rehabilitation purposes is considered very minimal. The use of high spatial resolution

images such as SPOT panchromatic image for road location as suggested by Williamson (1993) might be better.

6.2. Digital image analysis

6.2.1. Vegetated-bareland features

As with visual interpretation, dichotomized delineation of the area as vegetated and non-vegetated using the digital approach is highly possible. Forest could usually be classified apart from non-forest (Tuomisto *et al.* 1994). Even using lower spatial resolution Landsat MSS data, these two general land cover categories could be spectrally separated (e.g., Singh 1987). In this study, the adequate delineation of vegetated and non-vegetated areas is found more easily and faster using digital image analysis than pure visual interpretation techniques. The contrasting reflectance properties of bare areas and vegetation in the visible and infrared bands were better maximized by the image processing system than by using the naked eye on colour composites. The colour composite of TM-3, -4, and -5 was found to be most useful.

6.2.2. Forest type classes

Considering forest type, moderate to large overlaps in spectral values were observed between classes belonging to molave-dipterocarp forest, mangrove forest, coconut plantation, and mature tree plantation (figure 3). (All of these classes represent mature vegetation formation.) Only the classes of shrub and grass (young vegetation formation) were highly separated from this group. The visible band reflectance in these overlapping mature forest type classes (with mean values of 12.66, 14.07, 14.06, and 12.47, respectively) are significantly lower than the grass class (47.13) but close to shrub class (17.31). Likewise, their near-infrared and mid-infrared reflectance (mean values of 84.69, 74.63, 78.20, and 84.36 for near-infrared band) are lower than areas dominated by grasses and shrubs (94.88 and 132.93, respectively). This is possibly because of shadow or spectral traps (Mausel *et al.* 1993) present in these mature forest formation: characterized by its multi-layer canopy and more complex crown structure.

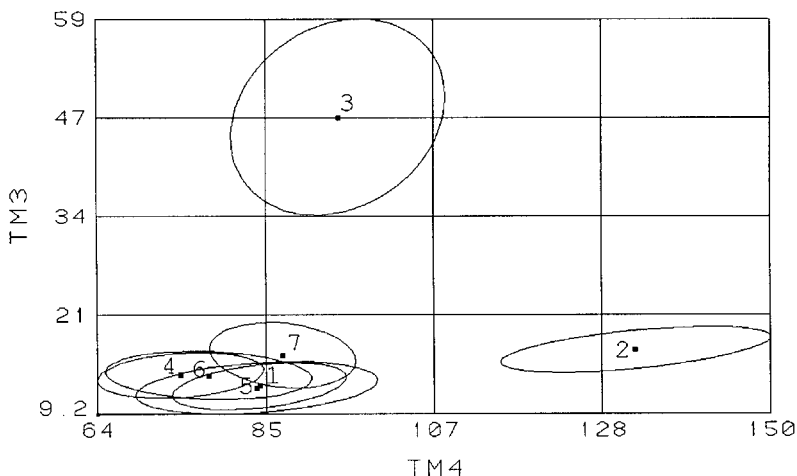


Figure 3. Ellipse plot of training samples for forest types (1 = molave-dipterocarp, 2 = shrub, 3 = grass, 4 = mangrove, 5 = tree plantation, 6 = coconut, 7 = mixed agri crops).

From the above statistics, adequate species-level or detailed forest type differentiation and mapping using digital techniques is not very promising. Forest types' spectrally overlapping classes suggest that substantial error in classification is inevitable even when the highly favored per-pixel classification algorithm such as maximum likelihood is used. The results of the supervised classification conducted here confirmed this: a number of molave-dipterocarp forest areas are incorrectly classified as coconut plantation or mangrove forest, and vice versa. For example, given the reference data, it is highly inaccurate for mangrove forests to yield more than 5000 hectares, especially on areas 700 m above mean sea level! Furthermore, obtaining a total area of more than 5000 ha for tree plantation is erroneous.

The results agree with the findings of Niemann (1993), Benson and deGloria (1985), and others, who found that digital classification of satellite multi-spectral data alone can not be used to determine species composition in forest stands. In their study of tropical secondary forest in Brazil, Lucas *et al.* (1993) also found similar difficulty because they could not adequately discriminate plantations (oil palm) from secondary forest using Landsat TM data.

On the other hand, this disagrees with the findings of Roy *et al.* (1991), Saxena *et al.* (1992), Hopkins *et al.* (1988), Karteris (1990), among others, who obtained relatively high classification accuracies in areas bearing detailed to semi-detailed segregation of classes. Explaining the difference would be very difficult because most of the interactions between the factors involved are complex. One of the most probable reasons is because tropical forest vegetation in the latter is divided into a few structurally clearly dissimilar types: the required level of classification detail has been rather low, and hence the general level of classification accuracy has been rather high (Tuomisto *et al.* 1994).

6.2.3. Crown cover and successional stages

Classes pertaining to crown cover have clearly no overlapping regions in the ellipse plot (figure 4), indicating their very high separability. In the near-infrared band, high crown cover class has the highest mean value (91.26) but the lowest in

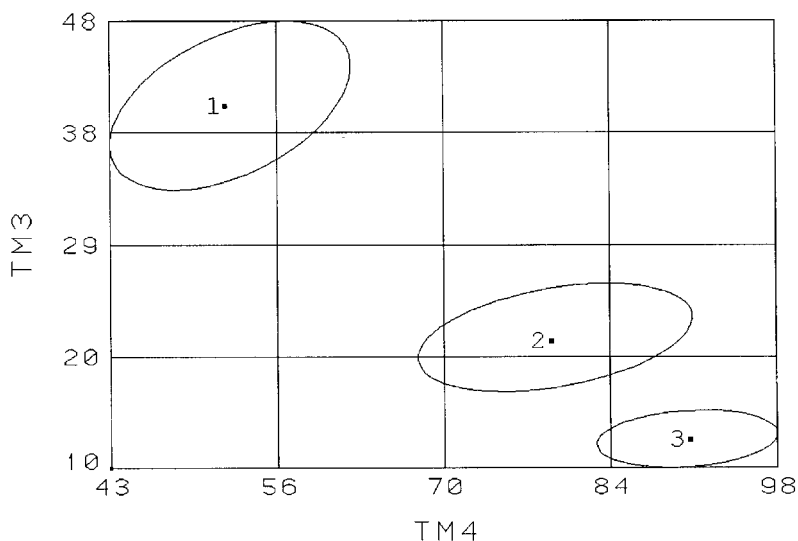


Figure 4. Ellipse plot of training samples for crown cover (1 = low, 2 = medium, 3 = high).

the red and mid-infrared bands (13.18 and 62.06, respectively). Low crown cover class has the highest reflectance in the red and mid-infrared bands (mean values of 41.14 and 116.55, respectively), but the lowest in the near-infrared band (53.09).

These observations for crown cover classes are similar in pattern with that of successional stage classes (figure 5). All classes have high spectral separability. The same as the low crown cover class, the initial succession stage class has the highest red and mid-infrared reflectance values (mean values of 38.85 and 139.64, respectively) but the lowest in the near-infrared bands (56.13). On the other hand, the advanced succession stage class has the highest near-infrared band (89.56) but has lowest red and mid-infrared bands (13.16 and 61.67, respectively).

These statistics indicate that land cover mapping featuring crown cover density is attainable in conditions that prevailed during this study. The spectral response of closed density forest is distinctly different from that of open and young shrublands, thus broad division of crown cover density (e.g., low, medium and high) is highly workable. This finding that crown cover density classes could be adequately mapped, generally agrees with that of JOFCA/ITTO (1993): a study of tropical forests in the Philippines and Thailand. In here, crown density classes were shown to be mapped using Landsat TM data at a very close class interval (i.e., 12 crown density classes: 0, 1–9, 10–19, 20–29 ..., 100 per cent).

One of the main reasons for the high spectral separability between crown classes in this study is the broad three-level categories (low, medium, and high) that were adopted and sampled. If increased to five, their spectral discrimination and their subsequent mapping would be difficult. (The associated problem is how to decide on class definition (e.g. for medium crown cover classes, is it 40–50 per cent or 40–60 per cent?), and to find the natural break, if any, of crown cover classes that could be captured by the sensor. Unlike forest types, subdivision of crown cover into classes is somewhat artificial. In the field, it is often easy to see and conceptually understand the natural groupings of tree species than crown cover classes.) However,

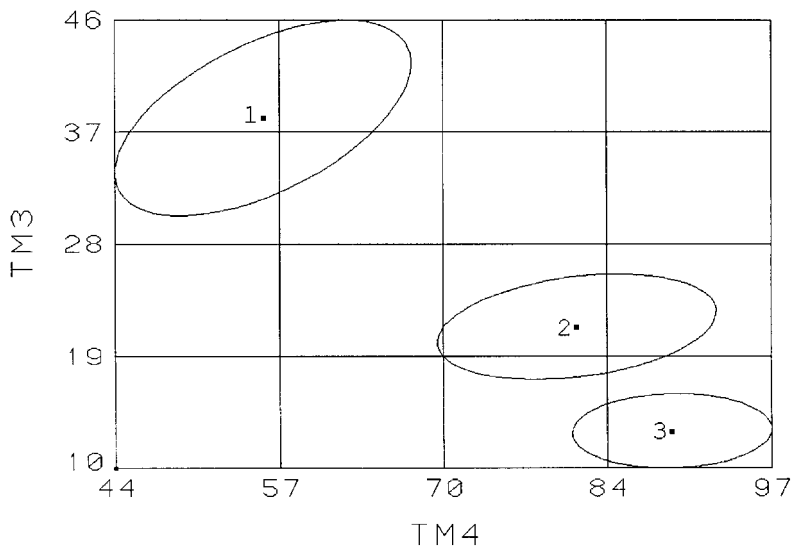


Figure 5. Ellipse plot of training samples for successional stages (1 = initial, 2 = intermediate, 3 = advanced).

for the objectives of most tropical forest rehabilitation planning and monitoring tasks, the broad three-level category of crown classes used in this study is sufficient (e.g., JOFCA/ITTO 1993, Deutsche Forst-Consult 1992).

Similar with crown cover, spectral separability of successional stage classes is high. This agrees with the results obtained by Mausel *et al.* (1993), where three levels of secondary succession forests were mapped accurately (from 82 to 88 per cent) using TM data of a region in Brazil. Again, this is possibly due to the broad level of successional stage categories that were used. The number and definition of categories of Mausel *et al.* (1993) are basically the same as the one used in this study. In general, the utility and limitations of TM data to map successional stages are the same as that of crown cover. It was demonstrated here that Landsat TM data has high potential utility in mapping crown cover and successional stage attributes which are essential in forest rehabilitation planning and monitoring.

6.2.4. Cultural and non-vegetated features

For non-vegetated classes, moderate overlap was observed between bare soil and built-up areas (which include road networks) (figure 6). Water and shadow have no overlap in their ellipses but their near-infrared and mid-infrared reflectance values are close (27.01, 32.62, and 29.34, 24.19, respectively). Both of these features have lower values in visible and infrared bands than any other surface features. As expected, the values of vegetated and bareland surface features are non-overlapping and are highly separable.

The supervised classification performed here confirmed that water bodies could not be mapped with adequate accuracy in an image containing rugged terrain areas. Many pixels are confused with topographic shadows in high relief areas since both have overlapping reflectance in the visible and reflective infrared bands. For instance, the area of 4396 hectares for water bodies is very far from reality. For cultural features such as road, trails, and human settlements, their digital-based mapping

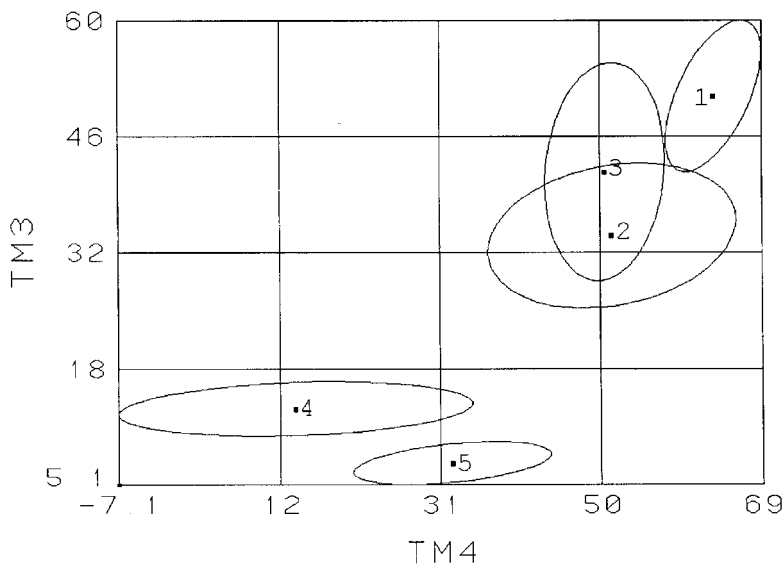


Figure 6. Ellipse plot of training samples for non-vegetated areas (1 = bare soil1, 2 = bare soil2, 3 = built-up, 4 = water, 5 = shadow).

from Landsat TM data is likewise not reliable. These man-made features have very close or overlapping spectral response with bare soil and other bareland areas. Similar confusion was observed by Beaubien (1994) in using a TM image in an area in Canada: some gravel roads are classified in the open land class. This suggests that Landsat TM data has limited utility in generating information for forest rehabilitation design.

6.3. *Unsupervised classification and its accuracy*

The clustering of the NDVI image yielded 96 spectral classes. The non-vegetated areas comprising bare soil, river beds, roads, rock outcrops, built-up areas, and others, were merged to form the 'bareland' categories. For vegetation, three categories were formed based on the percentage of crown cover: low crown cover (<10 per cent), medium crown cover (10–49 per cent), and high crown cover (50–100 per cent). Medium crown cover vegetation has the greatest area coverage (46.7 per cent of total area) while water is the lowest (0.9 per cent). The bareland areas, most likely candidate for rehabilitation, cover some 9063 ha (10.8 per cent).

Table 5 presents the error matrix for the classified NDVI image. Overall, the map accuracy is high i.e., 93.73 per cent, and a Kappa Index of Agreement (KIA) of a high 0.90 (a value of 1.00 indicates perfect association). The mapping accuracy is higher than the 85 per cent minimum accuracy criteria for tropical forest cover mapping suggested by Singh (1989). The producer's accuracy ranges from 81–96 per cent (while user's accuracy ranges from 86–95 per cent) for bareland, and low, medium, and high crown cover vegetation classes. The lower limits of these ranges belong to low crown cover vegetation class which has the lowest class accuracy. Water has an excellent accuracy of 100 per cent.

Almost all confusions were observed between the spectrally adjoining classes. Low crown cover vegetation class pixels were moderately confused with barelands and medium crown cover vegetation class pixels. Medium crown cover vegetation class exhibits minor confusion with low and high crown cover classes. On the other hand, high crown cover class is only confused with medium crown cover class in a

Table 5. Error matrix for the classified NDVI image.

Classified data	Reference data					RT	PROD %	USER %
	A	B	C	D	E			
A. Bareland	21	2	0	0	0	23	95.45	91.30
B. Low crown cover	1	26	3	0	0	30	81.25	86.67
C. Med crown cover	0	4	138	3	0	145	94.52	95.17
D. High crown cover	0	0	5	79	0	84	96.34	94.05
E. Water	0	0	0	0	5	5	100.00	100.00
CT	22	32	146	82	5	287		
Overall accuracy (%)								93.73
Kappa Index of Agreement (KIA)								0.90

RT is row total; CT is the column total; PROD is producer accuracy; USER is user accuracy

relatively low magnitude. The overall high classification accuracy could be attributed to the following reasons:

1. The number of crown cover categories used is relatively small, i.e., only three, which is sufficient for the objective of most forest rehabilitation planning. The classification accuracy declines when more classes are required because the differences among the classes becomes smaller (Tuomisto *et al.* 1994).
2. Inherent high spectral separability of bareland and vegetation cover.
3. The procedure conducted here provided a mechanism for thoughtful evaluation of each individual cluster before it is finally identified and merged.
4. The low-pass median filtering of the NDVI image helped in minimizing, if not removing possible noise.

The classification approach conducted here relies on the effectiveness of the NDVI's ability to enhance or reveal latent information where there is an inverse relation between spectral response to the biophysical phenomenon under survey (Campbell 1987, p. 384). In this study, these features are the bare soil and vegetated areas. However, the usual cautions must be observed when dealing with ratio-based images such as NDVI: the importance of data adjustments for additive factors such as atmospheric radiance (Crippen 1988), and the intensity blindness of ratio images (Lillesand and Kiefer 1987, p. 654).

Clustering, on the other hand, provided the procedure of examining the unknown pixels in an image and aggregating them into a number of classes based on the natural groupings or cluster present in the image values (Lillesand and Kiefer 1987, p. 685). Clustering in itself is not a subjective process, but the identification and merging of resulting spectral classes into information category is. In the whole procedure, the identification stage was found as the critical step in terms of the soundness and accuracy of classification outcomes. The accuracy of reference data consulted for this identification process must be high for best results. In addition, the classification method conducted here is considered semi-automatic: the identification and merging of spectral classes involved a large amount of analyst's intervention, especially if the number of classes is relatively large.

Utilizing NDVI image for land cover mapping will not yield information detailing the species or forest types. Instead, the information on crown cover density will be captured, including the vegetation-bareland delineation. Aside from NDVI, raw bands TM-3, -4 and -5 could be also used for this purpose. However, the presence of topographic shadows will be the major hindrance especially most forest landscapes in the tropics are located on high relief areas. In any case, crown cover density will be the most important and permissible information to extract in land cover mapping for forest rehabilitation purposes, given the current spatial and radiometric resolutions of sensors and the state of the art in normalizing topographic shadows.

It is undeniable that there are other classification methods that could be used to arrive at the same level of information what this study has obtained. However, of the methods tried, clustering of NDVI image was found to be simpler and more effective for this type of work, especially given the topographic shading. White (1995, pp. 141–142), in his review of current remote sensing approaches, reinforces NDVI's merit: *Unmixing approaches to forest inventory (Foody and Cox 1994) have been shown to produce comparable results to simpler techniques using the NDVI, suggesting that the added computational effort [of unmixing approaches] is not worthwhile in these cases (Shimabukuro et al. 1994).*

7. Summary and conclusions

The study sought to determine the details of land cover information that could be accurately mapped for forest rehabilitation planning by using contemporary satellite imagery, as well as the appropriate image processing techniques that could be routinely used. This was properly addressed and the following major findings are summarized below:

1. Dichotomized delineation of the area as vegetated and non-vegetated is highly possible using visual and digital image analysis. A colour composite image of TM-3, -4, and -5 in BGR, contrast stretched using histogram-equalization technique, is best for visual interpretation of the general land cover classes. The same TM bands are also favored for digital processing. However, topographic shadow is the major image-based constraint.

2. Adequate species-level or detailed forest-type differentiation and mapping under molave-dipterocarp forest formation is not permissible given the environmental condition, vegetation properties, and sensor's capability. Forest species composition is not spectrally distinct. The spectral responses of different species or forest types in the study area are not significantly different.

3. Classes of crown cover density and successional stage classes could be accurately mapped through (1) supervised classification of TM-3, -4, and -5, and (2) clustering of the NDVI image. The former is only true if topographic shadows in high relief areas will be accurately normalized.

4. Water features could not be digitally mapped with adequate accuracy in an image scene containing rugged terrain, because many of them are classified as topographic shadows. Using NDVI, their resulting values are similar or very close with bareland areas, thus they are not expected to be accurately classified and mapped.

5. Digital image-based mapping of cultural features such as road, trails, and human settlements from Landsat TM data is not reliable in rural environment like the one studied. Roads and trails traverse under forest canopy, while some residential areas are sparsely distributed and/or beneath the tree canopy. For mapping crown cover density in areas of rugged terrain, clustering of the NDVI image is more appropriate than the use of raw TM-3, -4, and -5.

It has been demonstrated that visual and digital analysis of Landsat TM data can yield land cover map useful for forest rehabilitation purposes. Crown cover density or canopy closure is regarded as the most useful and the most important detail of information Landsat TM data could provide in planning and monitoring of forest rehabilitation projects. Since the detailed level of species and forest type mapping was found unrealizable, and the delineation of water bodies and cultural features is not reliable under areas of mountainous terrain, satellite-based mapping of these features for forest rehabilitation planning is of less value. Thus, in mapping these features in rugged topography, ground survey or large-scale aerial photography are most likely the best alternatives at the present time. These findings and considerations, of course, are true under the current spatial and radiometric resolutions of sensors and the state of the art in normalizing topographic shadows.

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